

4. (a) Compounds are formed when atoms of different elements combine.

Draw **one** line from each compound to its formula.

[2]

Compound

Formula

carbon dioxide

CO

CO₂

Ca₂O

sodium hydroxide

NaOH

NaNO₃

Na₂CO₃

- (b) Magnesium oxide contains the ions Mg²⁺ and O²⁻.

Give the formula of magnesium oxide.

[1]

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- (c) Calcium sulfate has the formula CaSO_4 .

Calculate the relative formula mass (M_r) of calcium sulfate.

[2]

$$A_r(\text{Ca}) = 40$$

$$A_r(\text{S}) = 32$$

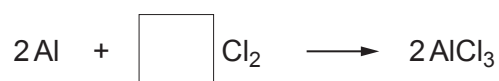
$$A_r(\text{O}) = 16$$

$$M_r = \dots\dots\dots$$

- (d) Aluminium chloride, AlCl_3 , is formed by the reaction of aluminium with chlorine.

Write a number in the box to balance the equation for this reaction.

[1]

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